

Lab Assignment 5

CS 331: Machine Learning

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Feb 4, 2025

In this lab you will perform classification on MNIST dataset. You may use sklearn to obtain the dataset, all other parts need to be coded from scratch. **You may work in groups already formed.**

During the lab hours you are expected to complete at least Qns 1-4. You are allowed to submit your complete solution by Feb 8, 11.59 pm.

1. Transform all features to binary using the following operation:
$$x_i^{new} = \mathbb{1}[x_i > 0]$$
2. Perform a split of the dataset into a training set $\mathcal{D}_{train}$ (80% of points) and test set $\mathcal{D}_{test}$ (remaining 20% of points).
3. Using $\mathcal{D}_{train}$ learn a Naive Bayes classifier with smoothing incorporated- use $a = b = 1$ for the hyper-parameters (smoothing).
4. Evaluate the classifier on $\mathcal{D}_{test}$ and report empirical 0-1 loss.
5. The values of a and b can greatly impact the performance. Use 5-fold cross validation to set the values of a and b and report the empirical 0-1 loss. You may use the following set of values for a and b : $\{1, 5, 10, 20, 80, 100, 1000\}$. So there are a total of $7 \times 7 = 49$ possible combinations for a and b .
6. Implement a K -nearest neighbour classifier and report the empirical 0-1 loss. For any input instance $\mathbf{x}$, the K -nearest neighbour algorithm looks at the K nearest neighbours of $\mathbf{x}$ from $\mathcal{D}_{train}$ and assigns the most frequently occurring label among these neighbours. Report empirical 0-1 loss on $\mathcal{D}_{test}$ for $K \in \{3, 5\}$.

Reference:

Duda, Hart and Stork, Pattern Classification, Chapter 4, Section 4.5.4

Submit all the code files and a pdf reporting the empirical 0-1 loss values.